



## MEQ

# Module Evaluation Survey

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# Least Squares, Tests

- Today: **Least Squares**: the math behind that call to fit.
- Wednesday: **Test Statistics** (Z, T, F/ ANOVA, LR, AIC, BIC,  $\chi^2$ )

This slide deck is only for Least Squares. I'm afraid the times when my lecture notes were up a week or two in advance are now soundly behind us...

# Intro, reminder of big picture

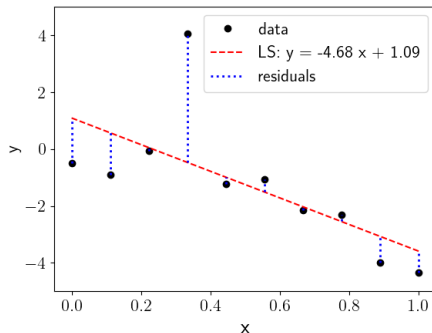
Fitting means estimating the parameters, and sometimes also the functional form, that best describes a dataset.

A functional form can have one parameter or lots of them.

The least squares approach minimises the residuals for all parameters simultaneously, to provide the best set of parameter estimates.

The math is a bit cumbersome, but I have done my best to keep it clean and clear.

# Least Squares



The Least Squares **fit** is a simultaneous estimation of the parameters of the function  $y(x)$ . The estimates are found by taking the derivatives of the sum of the squared **residuals** ( $\chi^2$ ) and setting them to zero.

# Chi Squared

Pearson's Chi Squared,  $\chi^2$ , is the sum of the squared **standardised residuals**:

$$\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

$y_i$  is the measurement of the RV  $Y$ .

$\mu$  is the True Value of  $y_i$ .

$\sigma_i$  is the standard deviation of a normal distribution centred on the true value  $\mu$ .

The numerator is the uncertainty in the model, aka the Residual Sum of Squares (RSS),

The denominator is the uncertainty in the data, aka the Total Sum of Squares (TSS).

# Why Chi Squared?

$$\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

Last week we wrote the Log Likelihood for Gaussian as:

$$\ln \mathcal{L}_\theta = -N \ln \sigma - N \ln \sqrt{2\pi} - \frac{1}{2} \sum_i^N \frac{(y_i - \mu)^2}{\sigma^2}$$

This assumes that  $\sigma$  is the same for every measurement  $y_i$ , or in other words that the  $y_i$  are from the same Gaussian distribution with true parameter  $\sigma$ .

If we avoid that assumption ([homoscedasticity](#)) we find:

$$\ln \mathcal{L}_\theta = - \sum_i^N \ln \sigma_i - N \ln \sqrt{2\pi} - \frac{1}{2} \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

Note that we no longer have two parameters to estimate, but  $N + 1$ ! **Oops...**

# Why Chi Squared?

$$\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

It's okay. All the  $\sigma_i$  estimates will have the same form:

$$\frac{\partial}{\partial \sigma_i} \ln \mathcal{L}_{\sigma_i} = - \sum_i^N \frac{1}{\sigma_i} - 0 + \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^3} = 0$$

therefore  $\sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^3} = \sum_i^N \frac{1}{\sigma_i}$  when  $\sum_i^N \sigma_i^2 = \sum_i^N (y_i - \mu)^2$

The best estimate for all  $\sigma_i$  is that which results in  $\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2} = 0$ .

This is also the case for the best estimate for  $\mu$ . Minimising  $\chi^2$  is exactly equivalent to finding the MLE for  $\mu$  and (all of the)  $\sigma$ .

## Chi Squared with Binned Data

$$\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

In real life we bin our measurements using histograms. We count the number of measurements in a small range of  $\Delta y$  (the bin width) and the height of the histogram bar is the number of measurements that fall in that interval.

In this case, we adjust our definition:

$$\chi^2 = \sum_i^N \frac{(n_i - \mu)^2}{n_i} = \sum_i^N \frac{(obs - exp)^2}{obs}$$

The use of  $n_i$  in place of  $\sigma_i^2$  is correct because in the binned case we are using bin counts, which means we are using **Poisson statistics**.

The Poisson RV is a rate (a **count**, in a given time),  $E[K] = \lambda$ , and the variance is  $V[K] = \lambda$ . For fixed time, the rate is a count, and the squared uncertainty  $V[K] = \sigma^2$  is  $\lambda \equiv n$ .

# The Equation of a Straight Line

A straight line in the  $x, y$  plane is described by  $y = mx + c$ , where  $m$  is the slope and  $c$  is the intercept on the  $y$ -axis.

The control variable is  $x$  and the RV is  $y$ : the thing we are measuring.

## Example:

Every hour, I measure my heart rate. Time is the control variable  $x$ , with evenly spaced intervals and negligible uncertainty. My measured heart rate is  $y$ , a random variable, with uncertainty.

If my heart rate increases steadily throughout the day from 70 to 100 beats per minute over a period of 10 hours, then  $m = 30/10 = 3$  and  $c = 70$ .  $y = 3x + 70$ .

# Constructing the $\chi^2$

$$y = mx + c$$

$$\chi^2 = \sum_i^N \frac{(y_i - \hat{\mu})^2}{\sigma_i^2} \quad \text{True } \mu \text{ is estimated from the fit: } \hat{\mu}.$$

A straight line fit of the form  $\hat{\mu} = \hat{m}x_i + \hat{c}$  provides  $\hat{m}$  and  $\hat{c}$ : the parameter estimates that minimise  $\chi^2$ . The Estimate for  $Y$  is the expected value of our measurement,  $\hat{\mu} = E[y_i]$ .

$$\chi^2 = \sum_i^N \frac{(y_i - (mx_i + c))^2}{\sigma_i^2}$$

$$\chi^2 = \sum_i^N \frac{1}{\sigma_i^2} (y_i^2 + m^2 x_i^2 + c^2 + 2cmx_i - 2cy_i - 2mx_i y_i)$$

# Derivatives of

$$\chi^2 = \sum_i^N \frac{1}{\sigma_i^2} (y_i^2 + m^2 x_i^2 + c^2 + 2cmx_i - 2cy_i - 2mx_i y_i)$$

As with the MLE, we find the parameters that minimise  $\chi^2$  by taking the first derivative with respect to each one.

We will put the [second derivatives](#) in our pocket for later, when we calculate the uncertainties.

$$\frac{\partial \chi^2}{\partial m} = \sum_i^N \frac{1}{\sigma_i^2} (2 m x_i^2 + 2 c x_i - 2 x_i y_i)$$

$$\frac{\partial^2 \chi^2}{\partial m^2} = \sum_i^N \frac{1}{\sigma_i^2} (2 x_i^2)$$

$$\frac{\partial \chi^2}{\partial c} = \sum_i^N \frac{1}{\sigma_i^2} (2 c + 2 m x_i - 2 y_i)$$

$$\frac{\partial^2 \chi^2}{\partial c^2} = \sum_i^N \frac{1}{\sigma_i^2} (2)$$

$$\frac{\partial^2 \chi^2}{\partial c \partial m} = \sum_i^N \frac{1}{\sigma_i^2} (2 x_i)$$

## Two parameter estimation

As with the MLE, we estimate the parameters by setting the partial derivatives to zero and solving for  $m$  and  $c$ , which gives us:

$$\sum_i \frac{1}{\sigma_i^2} y_i x_i = \sum_i \frac{1}{\sigma_i^2} (mx_i^2 + cx_i)$$

$$\sum_i \frac{1}{\sigma_i^2} y_i = \sum_i \frac{1}{\sigma_i^2} (mx_i + c)$$

We have found ourselves with two simultaneous equations (we have two parameters) and we need to solve them.

# Shorthand

Sums and squared sigmas are common everywhere, so we can tidy things up with shorthand:

$$s_1 = \sum_i^N \frac{1}{\sigma_i^2} \quad s_x = \sum_i^N \frac{x_i}{\sigma_i^2} \quad s_{xx} = \sum_i^N \frac{x_i^2}{\sigma_i^2} \quad s_y = \sum_i^N \frac{y_i}{\sigma_i^2} \quad s_{xy} = \sum_i^N \frac{x_i y_i}{\sigma_i^2}$$

We have:

$$\sum_i^N \frac{1}{\sigma_i^2} y_i x_i = \sum_i^N \frac{1}{\sigma_i^2} (mx_i^2 + cx_i) \Leftrightarrow s_{xy} = s_{xx} m + s_x c$$

$$\sum_i^N \frac{1}{\sigma_i^2} y_i = \sum_i^N \frac{1}{\sigma_i^2} (mx_i + c) \Leftrightarrow s_y = s_x m + s_1 c$$

# Solving simultaneous equations for $m$ and $c$

We have:

$$s_{xx} m + s_x c = s_{xy} \quad [1]$$

$$s_x m + s_1 c = s_y \quad [2]$$

Eliminate  $m$  by multiplying through and subtracting:

$$s_{xx}[2] - s_x[1]: \quad (s_{xx}s_1 - s_x s_x) c = s_{xx}s_y - s_x s_{xy}$$

$$c = \frac{s_{xx}s_y - s_x s_{xy}}{s_{xx}s_1 - s_x s_x}$$

Eliminate  $c$  by multiplying through and subtracting:

$$s_1[1] - s_x[2]: \quad (s_1 s_{xx} - s_x s_x) m = s_1 s_{xy} - s_x s_y$$

$$m = \frac{s_1 s_{xy} - s_x s_y}{s_1 s_{xx} - s_x s_x}$$

Note the denominator is  $s_1 s_{xx} - s_x s_x = \Delta$ : independent of  $y$ .

# Solving with Cramer's Method

A 'shortcut' to solving for  $m$  and  $c$  is to use Cramer's Method: we construct matrices of pairs of coefficients and take their determinants.

$$s_{xx} m + s_x c = s_{xy} \quad [1]$$

$$s_x m + s_1 c = s_y \quad [2]$$

The coefficients of  $m$  and  $c$ :  $\begin{vmatrix} s_{xx} & s_x \\ s_x & s_1 \end{vmatrix} = s_{xx}s_1 - s_x s_x = \Delta$

The coefficients of  $m$  and 1:  $c = \frac{\begin{vmatrix} s_{xx} & s_{xy} \\ s_x & s_y \end{vmatrix}}{\Delta} = (s_{xx}s_y - s_{xy}s_x)/\Delta$

The coefficients of 1 and  $c$ :  $m = \frac{\begin{vmatrix} s_{xy} & s_x \\ s_y & s_1 \end{vmatrix}}{\Delta} = (s_{xy}s_1 - s_x s_y)/\Delta$

Note the result is exactly the same as by solving directly. I don't find Cramer's way quicker...

# Solutions

The solutions for our two parameter estimates are:

$$\begin{bmatrix} \hat{m} \\ \hat{c} \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} s_{xy}s_1 - s_x s_y \\ s_{xx}s_y - s_{xy}s_x \end{bmatrix}$$

Where we have used the shorthand notation:

$$s_1 = \sum_i^N \frac{1}{\sigma_i^2} \quad s_x = \sum_i^N \frac{x_i}{\sigma_i^2} \quad s_{xx} = \sum_i^N \frac{x_i^2}{\sigma_i^2} \quad s_y = \sum_i^N \frac{y_i}{\sigma_i^2} \quad s_{xy} = \sum_i^N \frac{x_i y_i}{\sigma_i^2}$$

$$\text{And } \Delta = s_{xx}s_1 - s_x s_x$$

Note that without the shorthand notation we would be drowning in sums and squared sigmas. This is for the two parameters of a straight line with one independent variable.

# Parameter Uncertainties

Now, how to get the uncertainties on these monstrous solutions for the parameters  $m$  and  $c$ ?

Just as for the MLE, we can use a Taylor expansion of  $\chi^2$  about the estimated true parameter values:

$$\chi^2(m, c) = \chi^2(\hat{m}, \hat{c})$$

$$\begin{aligned}
 & + \cancel{\frac{\partial \chi^2}{\partial m} \bigg|_{\hat{m}}} (m - \hat{m}) + \cancel{\frac{\partial \chi^2}{\partial c} \bigg|_{\hat{c}}} (c - \hat{c}) && \text{Zero by construction} \\
 & + \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m^2} \bigg|_{\hat{m}} (m - \hat{m})^2 + \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial c^2} \bigg|_{\hat{c}} (c - \hat{c})^2 + 2 \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m \partial c} \bigg|_{\hat{m}, \hat{c}} (m - \hat{m})(c - \hat{c})
 \end{aligned}$$

The **second order partial derivatives** are our Information terms. These are the elements of the Curvature Matrix.

# Parameter Uncertainties

The Hessian (or Curvature) Matrix is:

$$H = \begin{bmatrix} \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m^2} & \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m \partial c} \\ \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m \partial c} & \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial c^2} \end{bmatrix} = \begin{bmatrix} s_{xx} & s_x \\ s_x & s_1 \end{bmatrix}$$

Where we have obtained the [nice simplified form](#) using the shorthand for the second derivatives we calculated earlier:

$$\frac{\partial^2 \chi^2}{\partial m^2} = \sum_i^N \frac{1}{\sigma_i^2} (2x_i^2) = 2s_{xx}$$

$$\frac{\partial^2 \chi^2}{\partial c^2} = \sum_i^N \frac{1}{\sigma_i^2} (2) = 2s_1$$

$$\frac{\partial^2 \chi^2}{\partial c \partial m} = \sum_i^N \frac{1}{\sigma_i^2} (2x_i) = 2s_x$$

# Parameter Uncertainties

$$\text{Inverse of Matrix: } \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{(ad - bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The **Inverse\*** of the Hessian Matrix is the **Covariance Matrix**  $\Sigma = H^{-1}$ . \*

$$H = \begin{bmatrix} s_{xx} & s_x \\ s_x & s_1 \end{bmatrix} \quad H^{-1} = \frac{1}{s_{xx}s_1 - s_x s_x} \begin{bmatrix} s_1 & -s_x \\ -s_x & s_{xx} \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} s_1 & -s_x \\ -s_x & s_{xx} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_m^2 & \text{cov}(m, c) \\ \text{cov}(m, c) & \sigma_c^2 \end{bmatrix}$$

We find:

$$\sigma_m^2 = \frac{s_1}{\Delta} \quad \sigma_c^2 = \frac{s_{xx}}{\Delta} \quad \text{cov}(m, c) = -\frac{s_x}{\Delta}$$

The nonzero covariance terms tell us the parameters are correlated. If we change the gradient of the fit,  $m$ , we will change the intercept,  $c$ .

\* Recall the inverse of the Information was the minimum variance bound

# Matrix Notation

The Taylor expansion for  $\chi^2$  about  $\theta = \hat{\theta}$  is:

$$\chi^2(m, c) = \chi^2(\hat{m}, \hat{c}) + \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m^2} \bigg|_{\hat{m}} (m - \hat{m})^2 + \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial c^2} \bigg|_{\hat{c}} (c - \hat{c})^2 + 2 \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m \partial c} \bigg|_{\hat{m}, \hat{c}} (m - \hat{m})(c - \hat{c})$$

In matrix notation this is:

$$\chi^2(m, c) - \chi^2(\hat{m}, \hat{c}) = \begin{bmatrix} m - \hat{m} \\ c - \hat{c} \end{bmatrix}^T \begin{bmatrix} \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m^2} & \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m \partial c} \\ \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial m \partial c} & \frac{1}{2!} \frac{\partial^2 \chi^2}{\partial c^2} \end{bmatrix} \begin{bmatrix} m - \hat{m} \\ c - \hat{c} \end{bmatrix} = \boxed{r^T H r}$$

And in Human this is:

Relative  $\chi^2 = \text{residuals vector}^T \cdot \text{Hessian} \cdot \text{residuals vector}$

# Matrix Notation

$$\chi^2_{rel} = \mathbf{r}^T \mathbf{H} \mathbf{r}$$

Any function that is linear in the parameters can be used in a residuals matrix, for example  $y = \theta_1 x^2 + \theta_2 x^3 + \theta_3 x^{-1}$  is linear in the parameters  $\theta$ .

We write  $y$  as a sum over  $\theta$  and  $f_x$ , where the function  $f_x$  does not have to be linear in  $x$ :  $y = \sum_j \theta_j f_{x,j}$  such that:

$$\chi^2_{rel} = (\mathbf{y} - \mathbf{f}_x \boldsymbol{\theta})^T \mathbf{H} (\mathbf{y} - \mathbf{f}_x \boldsymbol{\theta})$$

**Notes:**  $\mathbf{f}_x$  is a  $[Z \times N]$  matrix, with  $Z$  the number of parameters and  $N$  the number of measurements. The  $\mathbf{y}$  and  $\boldsymbol{\theta}$  are  $[N \times 1]$ .  $\mathbf{H}$  is  $[Z \times Z]$ .

# Matrix Notation

$$\chi_{rel}^2 = (y - f_x \theta)^T H (y - f_x \theta)$$

We minimise by taking the first derivatives with respect to our parameters  $\theta$  and using the **normal equations**:\*

$$\nabla_{\theta} (y - f_x \theta)^T (y - f_x \theta) = 2(f_x^T f_x \theta - f_x^T y)$$

$$\nabla_{\theta} \chi_{rel}^2 = 2(f_x^T H f_x \theta - f_x^T H y) = 0 \quad \therefore \quad (f_x^T H f_x) \theta = (f_x^T H) y$$

Multiply by the inverse of the LHS:

$$\theta = \left[ (f_x^T H f_x)^{-1} (f_x^T H) \right] y = B y$$

The covariance matrix in this case is

$$B H^{-1} B^T = (f_x^T H f_x)^{-1}$$

\*See a nice derivation from Eli Bendersky [here](#), or MIT version [here](#).

# Notes for reproducing with scipy

In the example above for a straight line, we kept  $s_1 = \sum_i^N \frac{1}{\sigma_i^2}$  free to vary across the  $i$  measurements.

If we were to assume constant variance for all our measurements, we would adjust our simultaneous equations accordingly, and our results would exactly match those of scipy's least squares implementation:

## Simultaneous Equations:

$$\sum x_i y_i = m \sum x_i^2 + c \sum x_i$$

$$\sum y_i = m \sum x_i + Nc$$

## Parameters:

$$\Delta = N \sum x_i^2 - (\sum x_i)^2$$

$$c = (\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i) / \Delta$$

$$m = (N \sum x_i y_i - \sum x_i \sum y_i) / \Delta$$

## Uncertainties:

$$\sigma_y^2 = \frac{1}{N-2} \sum (\hat{m}x_i + \hat{c} - y_i)^2$$

$$\sigma_c^2 = \sigma_y^2 \sum x_i^2 / \Delta$$

$$\sigma_m^2 = \sigma_y^2 N / \Delta$$

A potential 'gotcha' in doing this exercise is that the Covariance matrix returned by `least_squares` has been scaled. We need to scale it back using the reduced  $\chi^2/\nu$  - see python/workshop examples.

# Definitions

## MIT's excellent stats notes

Linear regression  
Class 25, 18.05  
Jeremy Orloff and Jonathan Bloom

### 1 Learning Goals

1. Be able to use the method of least squares to fit a line to bivariate data.
2. Be able to give a formula for the total squared error when fitting any type of curve to

3. Be able to say the words homoscedasticity and heteroscedasticity.

**Ordinary Least Squares (OLS):** For functions that are linear in their parameters.  
This does not mean they have to be linear in the RVs.

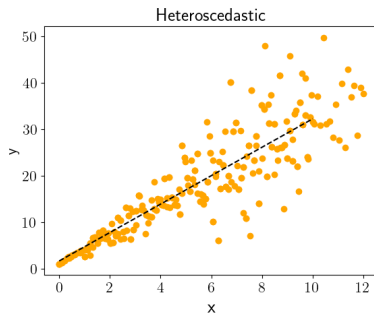
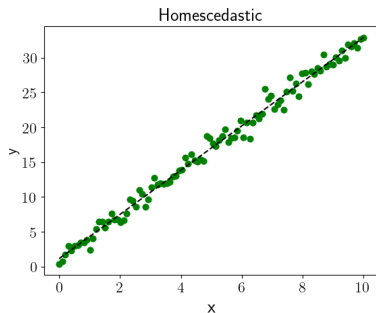
**Nonlinear Least Squares (OLS):** For functions that are nonlinear in their parameters.

If the uncertainties on the measured data are **homoscedastic**, and serially uncorrelated (linearly), then the parameter estimates of OLS are the Minimum Variance Unbiased Estimates (MVUE - see Gauss-Markov theorem).

If the data additionally have Gaussian uncertainties, then OLS and MLE are equivalent.

# Variance of Residuals

For OLS to be the MVUE, the data should be Homoscedastic.



If it is not, we can transform the data, or we can use Weighted LS (Like the Generalised LS example we went through, but with zero off-diagonal terms in the Cov matrix).

# Extreme Outliers

$$\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

Squaring the residuals gives large weight to outliers: values of  $y_i$  that deviate significantly from the best fit line that would result if they were excluded.

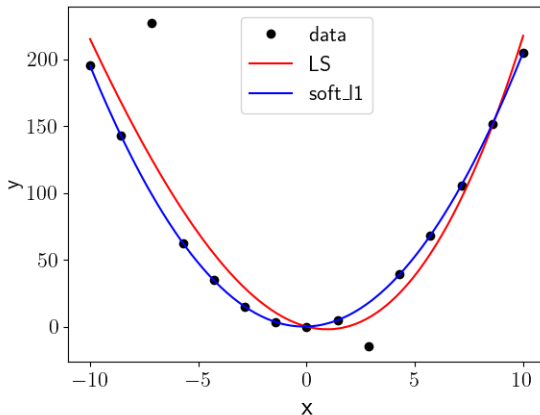
**Robust Least Squares** refers to using a different statistic in the numerator (one that is less sensitive to outliers).

One method for dealing with this is to use the Least Absolute Deviations method, which does not square the numerator.

In `scipy.optimize.least_squares`, this is controlled by specifying the `loss` argument - see workshop problems.

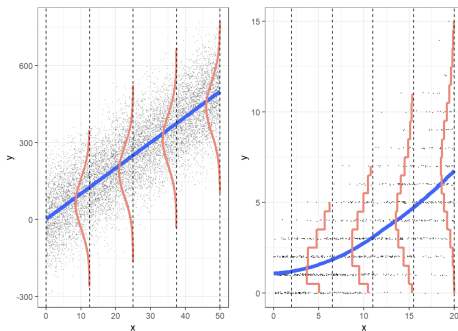
# Extreme Outliers

Example using `loss='soft_l1', f_scale=noise` to ensure the outliers do not throw off the fit.



# Other kinds of Regression

If the uncertainties on the data do not follow a normal distribution, we can use a Generalized Linear Model (GLM)- see python example with `statsmodels`.



PictureCredit: Roback & Legler

# Nonlinear Models

Functions that are not linear in the parameters, and cannot be linearised, can be solved iteratively rather than analytically. Methods for this include Simplex, and Gradient Search:

- 1 Build a  $N_p$ -Dimensional parameter space such that each point in space is defined by a unique set of  $\theta$ .
- 2 First derivative  $\nabla_{\theta_j} \chi^2$  indicates sign and magnitude of slope.
- 3 Go downwards: we are seeking the minimum.  $\theta_{j+1} = \theta_j - \gamma \nabla_{\theta} \chi^2$
- 4 Repeat.

Things can go wrong with this, of course. Getting stuck in local minima, hyper focusing, and missing the turning points are all associated with choosing the step size  $\gamma$  unwisely. Dark arts.

# Thanks For Your Time

# Fin.